



Implementation Plan: MVP Billing System (Manual Plan Tracking)

Overview

Objective: Implement a simple billing system to track organizations on Free, Standard, or Premium plans with manual plan management by the super admin. No automated payments yet—just plan tracking, limits, and manual upgrades.

Timeline: 3 weeks

Priority: High (required for monetization)

Dependencies: Authentication, Organization/Workspace management, Product activation system

Phase 1: Database Schema & Plan Configuration (Week 1)

User Story 1.1: Plan Definition

As the platform owner, I want to define subscription plans with prices and limits so I can track which organizations are on which tier.

Acceptance Criteria

- [] `plans` table created in Convex with fields: `key`, `name`, `monthlyPrice`, `annualPrice`, `currency`, `isActive`
- [] Three default plans seeded: free (₦0), standard (₦7,500/month), premium (₦20,000/month)
- [] Prices stored in kobo (smallest currency unit)
- [] Plans can be marked `isActive: false` to hide from new signups
- [] Admin can view all plans in dashboard
- [] Admin can update plan prices (affects new organizations only)

Technical Tasks

- [] Add `plans` table to `convex/schema.ts`
- [] Create indexes: `by_key`, `by_active`
- [] Seed script to create default plans on first deployment

- [] Admin API: GET /v1/admin/plans, PATCH /v1/admin/plans/:planKey
- [] Admin UI: Plans management page

User Story 1.2: Organization Subscription Tracking

As the platform owner, I want to track which plan each organization is on so I know who has access to what features.

Acceptance Criteria

- [] subscriptions table created with fields: workspaceId, planKey, status, currentPeriodStart, currentPeriodEnd, cancelAtPeriodEnd
- [] Every organization automatically gets a free plan subscription on creation
- [] Subscription status can be: active, cancelled, past_due
- [] Admin can see subscription details for any organization
- [] Admin can manually change an organization's plan
- [] Plan changes take effect immediately (no proration for MVP)

Technical Tasks

- [] Add subscriptions table to convex/schema.ts
- [] Create indexes: by_workspace, by_plan, by_status
- [] Modify organization creation to auto-create free subscription
- [] Admin API: GET /v1/admin/organizations/:workspaceId/subscription, POST /v1/admin/organizations/:workspaceId/subscription/change
- [] Admin UI: View subscription, change plan dropdown

User Story 1.3: Usage Tracking Foundation

As the platform owner, I want to track usage counters per organization so I can enforce plan limits.

Acceptance Criteria

- [] usageCounters table created with fields: workspaceId, featureKey, periodStart, periodEnd, usageValue, limitValue
- [] Track these counters per workspace:
 - workspace.count (number of workspaces owned by user)
 - apps.count (number of activated apps in workspace)
 - members.count (number of members in workspace)
 - products.count (number of products in workspace)

- ☐ Counters reset monthly (or on plan change)
- ☐ Admin can view usage for any organization

Technical Tasks

- ☐ Add `usageCounters` table to `convex/schema.ts`
- ☐ Create indexes: `by_workspace`, `by_workspace_feature`
- ☐ Create Convex queries to count resources per workspace
- ☐ Admin API: `GET /v1/admin/organizations/:workspaceId/usage`
- ☐ Admin UI: Usage dashboard with progress bars

Phase 2: Plan Limits & Enforcement (Week 2)

User Story 2.1: Plan Limits Configuration

As the platform owner, I want to define hard limits for each plan so I can prevent abuse of the free tier.

Acceptance Criteria

- ☐ Plan limits hardcoded in backend (not UI-configurable for MVP):

```
const PLAN_LIMITS = {
  free: {
    maxWorkspaces: 1,
    maxAppsPerWorkspace: 1,
    maxMembers: 2,
    maxProducts: 500,
    maxTransactions: 300,
  },
  standard: {
    maxWorkspaces: 3,
    maxAppsPerWorkspace: 3,
    maxMembers: 10,
    maxProducts: 5000,
    maxTransactions: 5000,
  },
  premium: {
    maxWorkspaces: 10,
    maxAppsPerWorkspace: 'unlimited',
    maxMembers: 50,
    maxProducts: 25000,
    maxTransactions: 25000,
  },
};
```

- ☐ Admin can view limits in documentation (not editable in UI yet)
- ☐ Limits checked before resource creation

Technical Tasks

- ☐ Create `backend/src/config/planLimits.ts` with hardcoded limits
- ☐ Create `backend/src/services/entitlementService.ts` to check limits
- ☐ Add limit-checking middleware to resource creation endpoints
- ☐ Return clear error messages when limit reached: "Free plan allows only 1 application. Upgrade to Standard for more."

User Story 2.2: Workspace Creation Limits

As a free user, I should be blocked from creating a second workspace so I understand the plan limitations.

Acceptance Criteria

- ☐ User on Free plan can own only 1 workspace
- ☐ Attempting to create 2nd workspace shows upgrade prompt
- ☐ User on Standard plan can own up to 3 workspaces
- ☐ User on Premium plan can own up to 10 workspaces
- ☐ Workspace creation endpoint checks limit before creating
- ☐ Error message: "Your Free plan includes 1 workspace. Upgrade to Standard to create more."

Technical Tasks

- ☐ Add `requireWorkspaceCreationEntitlement` middleware
- ☐ Query user's existing workspace count
- ☐ Compare against plan limit
- ☐ Return 403 with upgrade suggestion if limit reached

User Story 2.3: Application Activation Limits

As a free user, I should be blocked from activating a second application so I understand I need to upgrade.

Acceptance Criteria

- ☐ Free plan: can activate only 1 app per workspace
- ☐ Standard plan: can activate up to 3 apps per workspace
- ☐ Premium plan: can activate all available apps
- ☐ App activation endpoint checks limit
- ☐ App launcher shows "Activate" button disabled with tooltip when limit reached

- ☐ Error message: "Free plan includes 1 application. Upgrade to Standard to activate more."

Technical Tasks

- ☐ Add `requireAppActivationEntitlement` middleware
- ☐ Count active apps in workspace
- ☐ Check against plan limit
- ☐ Update frontend app launcher to show disabled state

User Story 2.4: Member Invitation Limits

As a free user, I should be blocked from inviting a 3rd member so I know I've hit the plan limit.

Acceptance Criteria

- ☐ Free plan: max 2 members per workspace
- ☐ Standard plan: max 10 members per workspace
- ☐ Premium plan: max 50 members per workspace
- ☐ Invitation endpoint checks member count
- ☐ Error message: "Free plan allows 2 members. Upgrade to Standard for more."
- ☐ Member list shows "2/2 members" with progress indicator

Technical Tasks

- ☐ Count workspace members before sending invitation
- ☐ Add `requireMemberInvitationEntitlement` middleware
- ☐ Update member list UI to show usage

User Story 2.5: Product Catalog Limits

As a free user, I should be blocked from adding more than 500 products so I understand the limitation.

Acceptance Criteria

- ☐ Free plan: max 500 products per workspace
- ☐ Standard plan: max 5,000 products per workspace
- ☐ Premium plan: max 25,000 products per workspace
- ☐ Product creation endpoint checks limit
- ☐ CSV import validates total product count
- ☐ Error message: "Free plan allows 500 products. Upgrade to Standard for more."

- ☐ Product list shows "487/500 products" with progress bar

Technical Tasks

- ☐ Count products in workspace before creation
- ☐ Add `requireProductCreationEntitlement` middleware
- ☐ Update product list UI to show usage
- ☐ CSV import validates against limit before processing

Phase 3: Admin Dashboard & Manual Management (Week 3)

User Story 3.1: Admin Plan Management

As the super admin, I want to view and update plan prices so I can adjust pricing as I validate with customers.

Acceptance Criteria

- ☐ Admin dashboard shows all plans with current prices
- ☐ Admin can update monthly and annual prices
- ☐ Price changes affect only new organizations (existing orgs keep old price)
- ☐ Admin can toggle `isActive` to hide/show plans
- ☐ Cannot delete plans that have active subscriptions
- ☐ Audit log records price changes

Technical Tasks

- ☐ Admin UI: Plans management page
- ☐ Form to edit `monthlyPrice`, `annualPrice`
- ☐ Toggle switch for `isActive`
- ☐ Backend validation: cannot delete active plans
- ☐ Audit log: `billing.plan_price_updated`

User Story 3.2: Admin Organization Subscription Management

As the super admin, I want to manually change an organization's plan so I can upgrade customers who contact me.

Acceptance Criteria

- ☐ Admin can view any organization's current plan
- ☐ Admin can change plan from Free → Standard → Premium
- ☐ Admin can downgrade plan (with warning)
- ☐ Plan change is immediate (no proration)
- ☐ Admin can set custom `currentPeriodEnd` date
- ☐ Admin can mark subscription as `cancelled` or `past_due`
- ☐ Organization receives notification on plan change

Technical Tasks

- ☐ Admin UI: Organization detail page with subscription section
- ☐ Dropdown to select new plan
- ☐ Date picker for `currentPeriodEnd`
- ☐ Status dropdown: `active`, `cancelled`, `past_due`
- ☐ Backend: POST `/v1/admin/organizations/:workspaceId/subscription/change`
- ☐ Send notification email on plan change

User Story 3.3: Admin Usage Dashboard

As the super admin, I want to see usage stats for any organization so I can help them when they hit limits.

Acceptance Criteria

- ☐ Admin can view usage dashboard for any workspace
- ☐ Shows: workspaces owned, apps activated, members, products, transactions
- ☐ Shows plan limits vs current usage (e.g., "3/10 members")
- ☐ Visual progress bars for each metric
- ☐ Admin can override limits for specific orgs (temporary)
- ☐ Admin can see which orgs are at >80% of any limit

Technical Tasks

- ☐ Admin UI: Usage dashboard page
- ☐ Query usage counters for workspace
- ☐ Display as progress bars with percentages
- ☐ "Override limit" button (temporary, expires in 30 days)
- ☐ Query: organizations with usage > 80% of limit

User Story 3.4: Admin Revenue Overview

As the super admin, I want to see basic revenue stats so I know how many organizations are paying.

Acceptance Criteria

- ☐ Dashboard shows: total organizations, organizations by plan (Free/Standard/Premium)
- ☐ Shows monthly recurring revenue (MRR) based on plan prices
- ☐ Shows organizations with `past_due` status
- ☐ Shows organizations cancelled in last 30 days
- ☐ Simple chart: plan distribution over time
- ☐ Export to CSV for manual reconciliation

Technical Tasks

- ☐ Admin UI: Revenue overview dashboard
- ☐ Aggregate queries: count by plan, sum of prices
- ☐ Filter: `status = 'past_due', cancelAtPeriodEnd = true`
- ☐ CSV export endpoint
- ☐ Simple bar chart (Chart.js or similar)

User Story 3.5: Manual Upgrade Flow

As a customer, I want to request a plan upgrade so I can move from Free to Standard when I'm ready.

Acceptance Criteria

- ☐ Organization settings shows current plan and limits
- ☐ "Request Upgrade" button visible on Free plan
- ☐ Clicking opens modal with: "Contact us on WhatsApp: +234-XXX-XXX-XXXX" or email
- ☐ Admin receives upgrade request notification
- ☐ Admin manually upgrades organization in dashboard
- ☐ Customer receives email confirmation: "Your organization is now on Standard plan"

Technical Tasks

- ☐ Organization settings UI: "Request Upgrade" button
- ☐ Modal with contact info
- ☐ Admin notification: email or in-app
- ☐ Admin workflow: receive request → upgrade in dashboard → send confirmation

- ☐ Email template: plan upgrade confirmation

Phase 4: Usage Tracking & Limit Enforcement (Ongoing)

User Story 4.1: Real-Time Usage Counters

As the system, I need to track usage in real-time so I can enforce limits accurately.

Acceptance Criteria

- ☐ Workspace count updated when user creates/joins workspace
- ☐ App count updated when app activated/deactivated
- ☐ Member count updated when member joins/leaves
- ☐ Product count updated on product create/archive
- ☐ Transaction count updated on each sale/purchase
- ☐ Counters reset on first day of month (or plan change)
- ☐ Usage queries are fast (indexed)

Technical Tasks

- ☐ Convex mutations to update counters on each action
- ☐ Monthly reset cron job (or Convex scheduled function)
- ☐ Indexes for fast counting
- ☐ Caching layer if needed for performance

User Story 4.2: Soft Limit Warnings

As a user, I want to be warned when I'm approaching my plan limit so I can upgrade before being blocked.

Acceptance Criteria

- ☐ Warning at 80% usage: "You've used 400/500 products"
- ☐ Warning at 90% usage: "You've used 450/500 products"
- ☐ Warning at 100% usage: "You've reached your limit. Upgrade to continue."
- ☐ Warnings shown in relevant UI (product list, member list, etc.)
- ☐ Email notification at 90% usage (optional for MVP)
- ☐ Warnings include upgrade CTA

Technical Tasks

- ☐ Calculate usage percentage on each resource list
- ☐ Show warning banner at 80%, 90%, 100%
- ☐ Email notification service (optional)
- ☐ Upgrade CTA links to upgrade request flow

User Story 4.3: Hard Limit Blocks

As the system, I need to block actions that exceed plan limits so I prevent abuse.

Acceptance Criteria

- ☐ Return 403 error when limit reached
- ☐ Error message includes: current usage, limit, upgrade suggestion
- ☐ Frontend disables action buttons when limit reached
- ☐ Audit log records limit-reached events
- ☐ Admin notified when org hits limit (for sales follow-up)

Technical Tasks

- ☐ Middleware returns 403 with structured error
- ☐ Frontend error handler shows upgrade prompt
- ☐ Audit log: `billing.limit_reached`
- ☐ Admin notification: daily digest of orgs at limit

Technical Implementation Details

Database Schema

```
// convex/schema.ts

plans: defineTable({
  key: v.string(), // "free", "standard", "premium"
  name: v.string(),
  monthlyPrice: v.number(), // kobo
  annualPrice: v.optional(v.number()), // kobo
  currency: v.string(), // "NGN"
  isActive: v.boolean(),
  createdAt: v.number(),
  updatedAt: v.number(),
})
.index("by_key", ["key"])
.index("by_active", ["isActive"]);
```

```

subscriptions: defineTable({
  workspaceId: v.id("workspaces"),
  planKey: v.string(),
  status: v.union(
    v.literal("active"),
    v.literal("cancelled"),
    v.literal("past_due")
  ),
  currentPeriodStart: v.number(),
  currentPeriodEnd: v.number(),
  cancelAtPeriodEnd: v.boolean(),
  createdAt: v.number(),
  updatedAt: v.number(),
})
.index("by_workspace", ["workspaceId"])
.index("by_plan", ["planKey"])
.index("by_status", ["status"]);

usageCounters: defineTable({
  workspaceId: v.id("workspaces"),
  featureKey: v.string(),
  periodStart: v.number(),
  periodEnd: v.number(),
  usageValue: v.number(),
  limitValue: v.number(),
  updatedAt: v.number(),
})
.index("by_workspace", ["workspaceId"])
.index("by_workspace_feature", ["workspaceId", "featureKey"]);

```

API Endpoints

```

// Admin routes
GET    /v1/admin/plans
PATCH /v1/admin/plans/:planKey
GET    /v1/admin/organizations/:workspaceId/subscription
POST   /v1/admin/organizations/:workspaceId/subscription/change
GET    /v1/admin/organizations/:workspaceId/usage
GET    /v1/admin/revenue/overview

// User-facing routes
GET    /v1/workspaces/:workspaceId/subscription
GET    /v1/workspaces/:workspaceId/usage
POST   /v1/workspaces/:workspaceId/subscription/request-upgrade

```

Convex Queries & Mutations

```

// convex/plans.ts
export const list = query(...);
export const getByKey = query(...);
export const update = mutation(...);

// convex/subscriptions.ts

```

```
export const getByWorkspace = query(...);
export const updatePlan = mutation(...);
export const createDefault = mutation(...);

// convex/usageCounters.ts
export const getByWorkspace = query(...);
export const increment = mutation(...);
export const resetMonthly = mutation(...);
```

Testing Checklist

Unit Tests

- ☐ Plan limits enforced correctly
- ☐ Usage counters increment/decrement properly
- ☐ Subscription status changes work
- ☐ Admin can change plans
- ☐ Free plan defaults on org creation

Integration Tests

- ☐ Creating 2nd workspace on Free plan fails
- ☐ Activating 2nd app on Free plan fails
- ☐ Inviting 3rd member on Free plan fails
- ☐ Adding 501st product on Free plan fails
- ☐ Upgrading to Standard removes limits

E2E Tests

- ☐ Admin changes plan → user sees new limits immediately
- ☐ Usage dashboard shows correct counts
- ☐ Revenue overview calculates MRR correctly
- ☐ Upgrade request flow works end-to-end

Deployment Plan

Week 1: Schema & Tracking

- ☐ Deploy plans, subscriptions, usageCounters tables
- ☐ Seed default plans
- ☐ Deploy admin plan management UI
- ☐ Test: can view/edit plans

Week 2: Limits & Enforcement

- ☐ Deploy entitlement service
- ☐ Add middleware to resource creation endpoints
- ☐ Deploy usage tracking
- ☐ Test: limits enforced correctly

Week 3: Admin Dashboard

- ☐ Deploy admin subscription management
- ☐ Deploy usage dashboard
- ☐ Deploy revenue overview
- ☐ Test: manual upgrade flow

Success Metrics

- ✓ All organizations have a subscription record
- ✓ Plan limits enforced (workspace, apps, members, products)
- ✓ Admin can change plans manually
- ✓ Usage counters accurate within 5%
- ✓ Revenue dashboard shows correct MRR
- ✓ Upgrade request flow works (customer → admin → upgrade)

Out of Scope for MVP

- ✗ Automated payment collection (Paystack/Flutterwave)
- ✗ Self-service plan upgrades
- ✗ Proration calculations
- ✗ Invoice generation
- ✗ Payment failure handling
- ✗ Grace period automation
- ✗ Dunning management
- ✗ Tax calculations
- ✗ Multi-currency support (NGN only)

This implementation plan is ready for development. All user stories must be completed for MVP billing to be considered functional.